

Migration Rehearsal Agent

Evidence before a PostgreSQL schema change

01 The problem

Database migrations can succeed while application queries, dependent objects, or workflows break. A safe rehearsal exposes downstream effects before a live schema change and gives reviewers concrete evidence about affected code and database objects; successful DDL alone is not enough.

02 What the agent reaches

For the fixed `orders.status` → `order_status` rename, it scans GitHub source, records file-and-line findings and scan coverage, inventories database dependencies, copies 300 users and 500 orders into staging, applies the rename there, and checks direct SQL and the cloned view separately. The old direct query fails; PostgreSQL preserves the view output named `status`. A sandbox script produces a structured evidence report.

03 Where it stops

TrueForge pauses before `postgres-production.execute_sql`. Only a human approval applies the rename to `demo.public.orders`. Denying leaves `public` unchanged. The gate still appears when the rehearsal finds breakage; the report recommends fixing direct references first, and approving that risk remains the reviewer's decision.

04 Architecture

GitHub MCP → code scan
PostgreSQL MCP → staging proof
TrueForge sandbox → JSON report
Approval gate → demo public schema

Two database aliases share one MCP server but carry separate approval policies.

05 How TrueForge is used

TrueForge runs the agent/tool loop, authenticates GitHub, bridges MCP calls into Code Mode, executes the report script, and enforces the production approval gate. Connector credentials remain in the harness, outside the sandbox. The agent manifest supplies the sequence and reporting rules.

06 Real versus mocked

Real: PostgreSQL, MCP calls, GitHub file reads, sandbox execution, and human approval. Mocked: seeded rows, disposable staging, sample source repository, and a public schema standing in for production. No live app traffic is exercised. The staging copy omits indexes and constraints; bounded scanning can miss dynamic or external SQL. Rollback is not automated.